Visual Instruction Tuning

LLaVA

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Contents

Instruction Tuning

- Google의 FLAN(Finetuned Language Models are Zero-Shot Learners) 논문에서 제안됨
- LLM 모델을 Instruction 데이터셋을 통해 fine-tuning을 진행하여 zero-shot 성능을 높이는 방법론

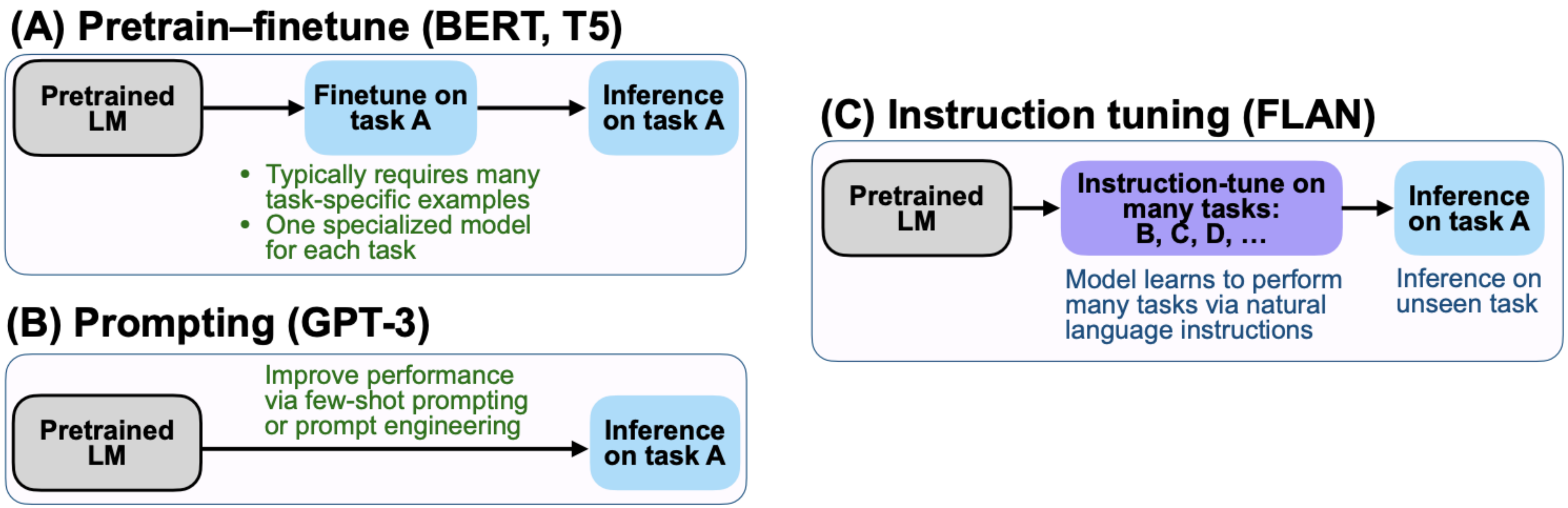
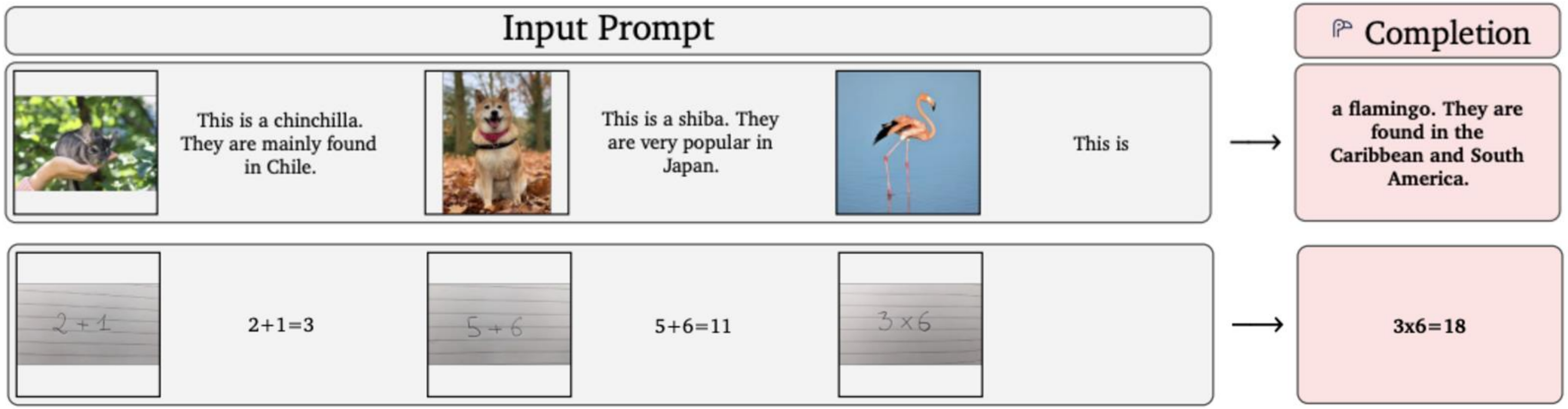


Figure 2: Comparing instruction tuning with pretrain–finetune and prompting.

출처 : https://arxiv.org/abs/2109.01652

Instruction Tuning

- OpenFlamingo , LLaMA-Adapter :
  오픈 소스 LLM인 LLaMA가 이미지 입력을 사용할 수 있도록 하여 multimodal LLM 구축
- promising task transfer 일반화 성능을 보임
- 일반적으로 language-only task에 비해 multimodal task에서는 성능이 좋지 못함

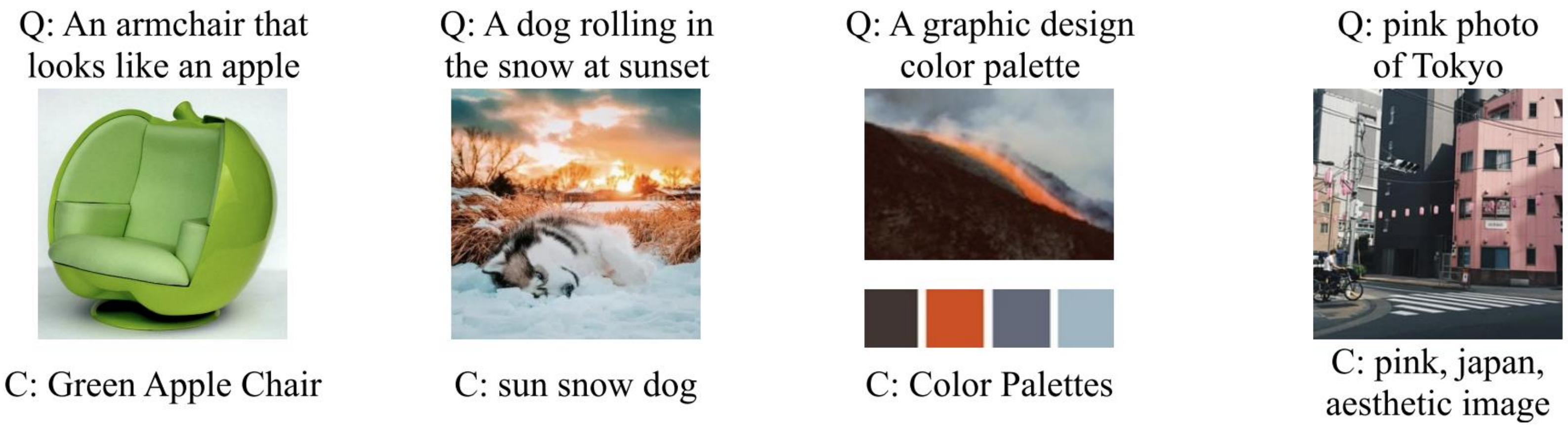


< Flamingo: a Visual Language Model for Few-Shot Learning >

출처 : https://storage.googleapis.com/deepmind-media/DeepMind.com/Blog/tackling-multiple-tasks-with-a-single-visual-language-model/flamingo.pdf

Visual Instruction Data Generation

- CC, LAION 등의 기존 multimodal dataset은 단순한 image-text pair data
- multimodal instruction-following data 구축 필요
  - 생성 프로세스 큰 시간 비용, human crowd sourcing 시 데이터가 잘 정의되지 않을 수 있어 ChatGPT/ language-only GPT4를 이용



< LAION-5B examples >

Visual Instruction Data Generation

$$\texttt{Human}: \mathbf{X}_{\mathrm{q}}\,\mathbf{X}_{\mathrm{v}}\ \texttt{<STOP>}\quad \texttt{Assistant}: \mathbf{X}_{\mathrm{c}}\ \texttt{<STOP>}$$

X_v (Image) , X_q (Question) , X_c (Caption)

- 적은 비용으로 데이터 생성
- Instructions와 response 모두에서 다양성과 심층적 추론 부족

- "Describe the image concisely."
- "Provide a brief description of the given image."
- "Offer a succinct explanation of the picture presented."
- "Summarize the visual content of the image."
- "Give a short and clear explanation of the subsequent image."
- "Share a concise interpretation of the image provided."
- "Present a compact description of the photo's key features."
- "Relay a brief, clear account of the picture shown."
- "Render a clear and concise summary of the photo."
- "Write a terse but informative summary of the picture."
- "Create a compact narrative representing the image presented."

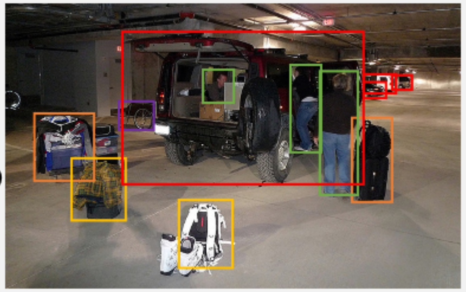
< The list of instructions for brief image description >

Visual Instruction Data Generation

- symbolic representation
  - visual content가 포함된 instruction-following data 생성하기 위함
  - Captions : 이미지에 대한 다양한 시각 개념 추출 및 설명
  - Bounding boxes : 각 물체 및 개념들의 위치와 정보 설명
  - language-only GPT를 위해 image 자체를 input으로 사용하지 않음

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

< captions & bounding boxes example>

Visual Instruction Data Generation

- 3 types instruction following data
  - instruction-following data 생성하기 위함
  - COCO dataset 사용하여 생성
  - Conversation , Deatailed description, Complex reasoning 3가지 타입으로 생성
  - 각 타입에 대해 사람이 직접 몇 개의 예시를 설계하여 사용
    - 데이터 수집 과정 중 유일한 human annotation
    - in-context learning 에서 GPT4 쿼리를 주기위한 seed example로 사용
    - assistant로 ChatGPT/GPT4로 실험해본 결과, GPT4가 더 좋은 품질의 data 생성함

Visual Instruction Data Generation

- 3 types instruction following data : Conversation
  - assistant가 이미지를 보고 human의 질문에 답하는 듯한 어조로 답변
  - 이미지의 시각적 내용들에 대해 다양한 질문들이 제기 됨
    (물체 유형, 개수, 움직임, 위치, 상대적 위치)
  - 분명한 답변이 가능한 질문만 고려됨
  - 58K 생성

Response type 1: conversation
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV).
Question: Where is the vehicle parked?
Answer: The vehicle is parked in an underground parking area, likely in a public garage.
Question: What are the people in the image doing?
Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Visual Instruction Data Generation

- 3 types instruction following data : Detailed Description
  - 이미지에 대한 풍부하고 포괄적인 설명을 포함하기 위해 질문 목록 직접 작성
  - 각 이미지마다 하나의 질문을 무작위로 골라서 GPT4에 물어보고 자세한 설명을 만들어내도록 함
  - 23K 생성

- "Describe the following image in detail"
- "Provide a detailed description of the given image"
- "Give an elaborate explanation of the image you see"
- "Share a comprehensive rundown of the presented image"
- "Offer a thorough analysis of the image"
- "Explain the various aspects of the image before you"
- "Clarify the contents of the displayed image with great detail"
- "Characterize the image using a well-detailed description"
- "Break down the elements of the image in a detailed manner"
- "Walk through the important details of the image"
- "Portray the image with a rich, descriptive narrative"
- "Narrate the contents of the image with precision"
- "Analyze the image in a comprehensive and detailed manner"
- "Illustrate the image through a descriptive explanation"
- "Examine the image closely and share its details"
- "Write an exhaustive depiction of the given image"

< The list of instructions for detailed image description >

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.

In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.

Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Visual Instruction Data Generation

- 3 types instruction following data : Complex Reasoning
  - 앞선 2가지 유형을 기반으로 심층적인 추론 질문 생성
  - 일반적으로 엄격한 논리를 따르는 단계별 추론 과정을 거침
  - 77K 생성

Response type 3: complex reasoning
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

Visual Instruction Tuning

Architecture

- image encoder : pre-trained CLIP 중 ViT-L/14

$$\mathbf{H}_{\mathrm{v}} = \mathbf{W} \cdot \mathbf{Z}_{\mathrm{v}}$$

$$\mathbf{Z}_{\mathrm{v}} = g(\mathbf{X}_{\mathrm{v}})$$

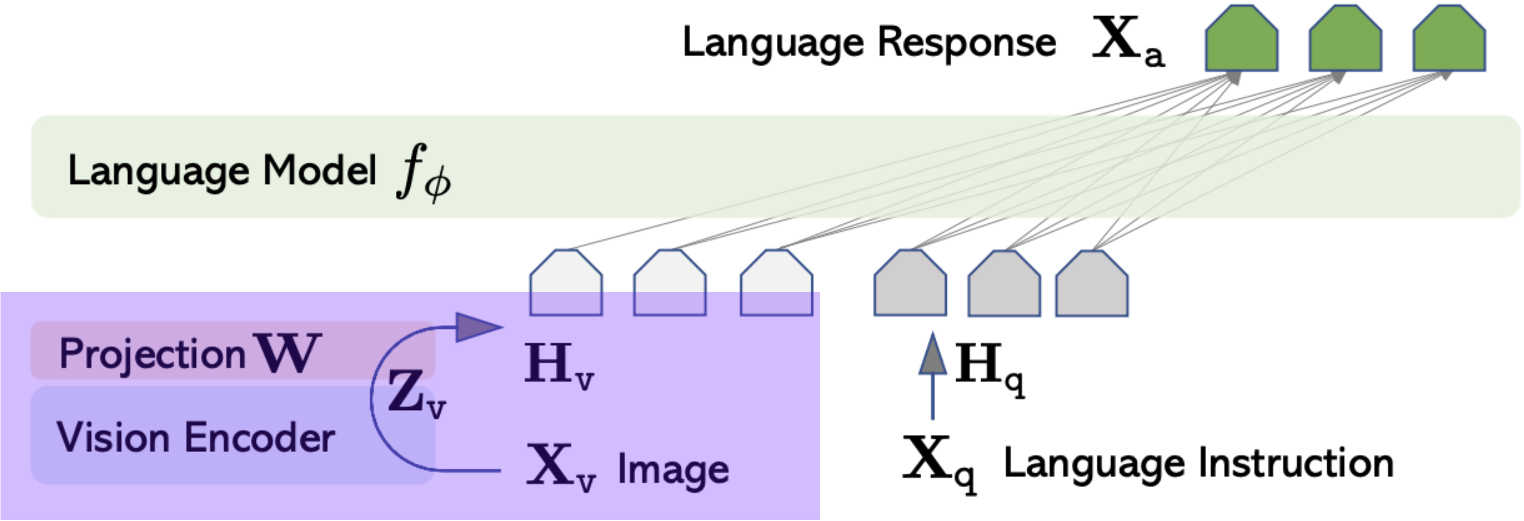


Figure 1: LLaVA network architecture.

Visual Instruction Tuning

Architecture

- Language model : Vicuna

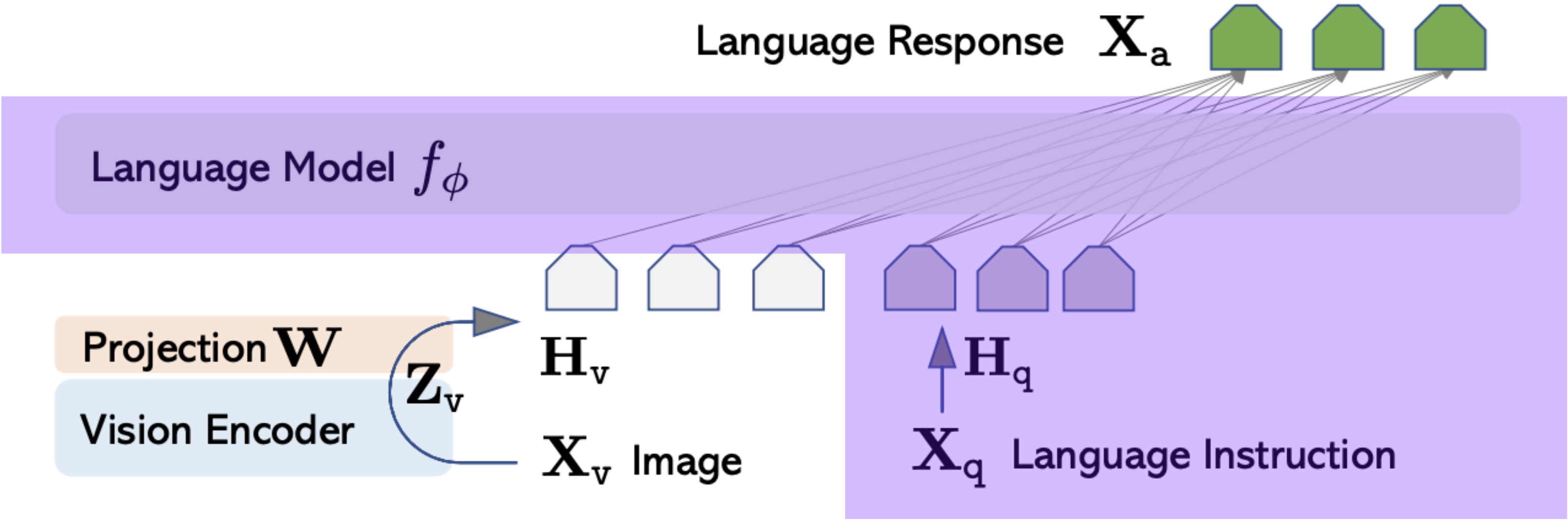


Figure 1: LLaVA network architecture.

Visual Instruction Tuning

Training

- 각 이미지 $\mathbf{X}_{\mathtt{v}}$ 에 대해서
  T개의 multiturn conversation data $(\mathbf{X}_{\mathtt{q}}^1, \mathbf{X}_{\mathtt{a}}^1, \cdots, \mathbf{X}_{\mathtt{q}}^T, \mathbf{X}_{\mathtt{a}}^T)$ 생성

- t번째 instruction

$$\mathbf{X}_{\mathtt{instruct}}^t = \begin{cases} \text{Randomly choose } [\mathbf{X}_{\mathtt{q}}^1, \mathbf{X}_{\mathtt{v}}] \text{ or } [\mathbf{X}_{\mathtt{v}}, \mathbf{X}_{\mathtt{q}}^1], & \text{the first turn } t = 1 \\ \mathbf{X}_{\mathtt{q}}^t, & \text{the remaining turns } t > 1 \end{cases}$$

- 일관된 multimodal instruction-following sequence 구성 가능

$\mathbf{X}_{\mathtt{system\text{-}message}}$ <STOP>
Human : $\mathbf{X}_{\mathtt{instruct}}^1$ <STOP> Assistant: $\mathbf{X}_{\mathtt{a}}^1$ <STOP>
Human : $\mathbf{X}_{\mathtt{instruct}}^2$ <STOP> Assistant: $\mathbf{X}_{\mathtt{a}}^2$ <STOP> $\cdots$

Visual Instruction Tuning

Training

- auto-regressive training objective 사용
- L개의 sequence에 대해 $\mathbf{X}_{\mathtt{a}}$ 의 확률 계산

$$p(\mathbf{X}_{\mathtt{a}}|\mathbf{X}_{\mathtt{v}}, \mathbf{X}_{\mathtt{instruct}}) = \prod_{i=1}^{L} p_{\boldsymbol{\theta}}(\boldsymbol{x}_i|\mathbf{X}_{\mathtt{v}}, \mathbf{X}_{\mathtt{instruct},<i}, \mathbf{X}_{\mathtt{a},<i})$$

$\mathbf{X}_{\mathtt{v}}$: image feature

$\mathbf{X}_{\mathtt{instruct},<i}$: 현재 예측 토큰 이전의 모든 instruction

$\mathbf{X}_{\mathtt{a},<i}$: 현재 예측 토큰 이전의 모든 answer

Visual Instruction Tuning

Training

- loss 계산시 예측하고자 하는 sequence/ token 만을 이용

$\mathbf{X}_{\texttt{system-message}}$ <STOP>
Human : $\mathbf{X}^1_{\texttt{instruct}}$ <STOP> Assistant: $\mathbf{X}^1_{\texttt{a}}$ <STOP>
Human : $\mathbf{X}^2_{\texttt{instruct}}$ <STOP> Assistant: $\mathbf{X}^2_{\texttt{a}}$ <STOP> $\cdots$

$$p(\mathbf{X}_{\texttt{a}}|\mathbf{X}_{\texttt{v}}, \mathbf{X}_{\texttt{instruct}}) = \prod_{i=1}^{L} p_{\boldsymbol{\theta}}(\boldsymbol{x}_i|\mathbf{X}_{\texttt{v}}, \mathbf{X}_{\texttt{instruct},<i}, \mathbf{X}_{\texttt{a},<i})$$

Visual Instruction Tuning

Training

- Stage 1: Pre-training for Feature Alignment
  - CC3M의 데이터를 595k의 image-text pair로 필터링
  - prompt 랜덤 샘플링하여 instruction tuning 수행
  - $\mathbf{W}$ 만 학습 수행 : visual tokenizer 학습

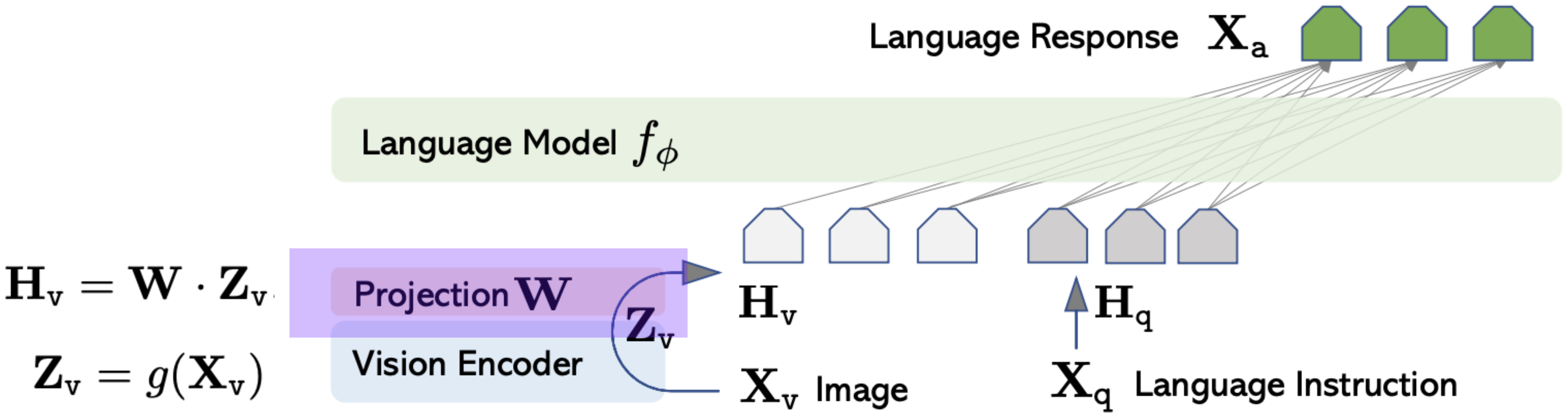


Figure 1: LLaVA network architecture.

Visual Instruction Tuning

Training

- Stage 2. Fine-tuning End-to-End
  - LLM과 사전학습한 projection layer 가중치 업데이트
  - Multimodal Chatbot :
    - 58k의 고품질 데이터를 사용하여 Chatbot 구축
    - 멀티턴과 싱글턴 대화 데이터를 고르게 샘플링하여 학습
  - Science QA :
    - 답변을 상세한 강의 및 설명과 함께 주석으로 달아둔 최초의 대규모 다중 모달 과학 질문 데이터셋
    - 각 질문은 자연어 또는 이미지 형태의 문맥(Context)과 함께 제공
    - 에이전트는 자연어로 추론 과정을 제공하고, 여러 선택지 중에서 답을 선택
    - 싱글턴 대화 데이터로 학습

Experiments & Conclusion

Multimodal Chatbot

- LLaVA는 비교적 적은
  multimodal instruction-following dataset으로 학습했음에도
  Multimodal GPT-4와 유사한 성능 확인
  Out of domain 이미지를 입력해도 해당 scene을 잘 이해함

- BLIP-2나 OpenFlamingo는 이미지를 묘사하는데 치중됨
  또한 User의 Insturction을 잘 따르지 않음

Visual input example, Extreme Ironing:

Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg

User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.
	[Start a new conversation, and clear the history]
User	What's happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car. The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4 [36]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	What is unusual about this image?
BLIP-2	a man is sitting on the back of a yellow cab
User	What is unusual about this image?
OpenFlamingo	The man is drying his clothes on the hood of his car.

Experiments & Conclusion

Multimodal Chatbot

LLaVA-Bench (In-the-Wild)

	Conversation	Detail description	Complex reasoning	All
OpenFlamingo [5]	19.3 ± 0.5	19.0 ± 0.5	19.1 ± 0.7	19.1 ± 0.4
BLIP-2 [28]	54.6 ± 1.4	29.1 ± 1.2	32.9 ± 0.7	38.1 ± 1.0
LLaVA	57.3 ± 1.9	52.5 ± 6.3	81.7 ± 1.8	67.3 ± 2.0
LLaVA†	58.8 ± 0.6	49.2 ± 0.8	81.4 ± 0.3	66.7 ± 0.3

- Challenging한 task와 일반화 성능을 확인하여 모델의 성능 평가
- BLIP-2 대비 29%, OpenFlamingo 대비 48%성능 향상을 확인
- 특히 이미지에 대해 심층적으로 답변하는 Complex reasoning 능력 우수

Experiments & Conclusion

Limitations

Challenging examples from LLaVA-Bench (In-the-Wild):

ICHIRAN Ramen [source]

Filled fridge [source]

	ICHIRAN Ramen [source]	Filled fridge [source]
Annotation	A close-up photo of a meal at ICHIRAN. The chashu ramen bowl with a spoon is placed in the center. The ramen is seasoned with chili sauce, chopped scallions, and served with two pieces of chashu. Chopsticks are placed to the right of the bowl, still in their paper wrap, not yet opened. The ramen is also served with nori on the left. On top, from left to right, the following sides are served: a bowl of orange spice (possibly garlic sauce), a plate of smoke-flavored stewed pork with chopped scallions, and a cup of matcha green tea.	An open refrigerator filled with a variety of food items. In the left part of the compartment, towards the front, there is a plastic box of strawberries with a small bag of baby carrots on top. Towards the back, there is a stack of sauce containers. In the middle part of the compartment, towards the front, there is a green plastic box, and there is an unidentified plastic bag placed on it. Towards the back, there is a carton of milk. In the right part of the compartment, towards the front, there is a box of blueberries with three yogurts stacked on top. The large bottle of yogurt is Fage non-fat yogurt, and one of the smaller cups is Fage blueberry yogurt. The brand and flavor of the other smaller cup are unknown. Towards the back, there is a container with an unknown content.
Question 1	What's the name of the restaurant?	What is the brand of the blueberry-flavored yogurt?
Question 2	Describe this photo in detail.	Is there strawberry-flavored yogurt in the fridge?

Experiments & Conclusion

Discussion & Conclusion

1. end-to-end 방식으로 작업별 조건을 학습하여,
   사전 학습된 diffusion model에 조건부 제어를 유연하게 할 수 있는
   neural network 제안

2. zero-convolution는 0의 초기값에서 점진적으로 최적화되기 때문에
   학습에 유해한 노이즈는 반영하지 않으면서
   각 작업별 조건에 최적화할 수 있음
   또한 새로운 레이어를 학습시키는 것과 비교하여,
   diffusion model의 fine tuning 속도보다 빠름

3. 이미지 생성에 대한 가중치들이 보존되기 때문에
   다양한 규모의 데이터셋에 강력함